



CMOS readout circuit design for infrared image sensors

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ABSTRACT

The infrared imaging system has been developed for more than 50 years, from the early stage the scanned imaging system using single unit detector to imaging system using focal plane detector arrays. For focal plane array detectors, the readout circuit is used to read the photon detector signal out. Charge coupled device had been used for the readout of the focal plane array detectors and currently CMOS technology is used. In this paper, readout circuit design using CMOS technology for infrared focal plane array detectors is reviewed. As an interface between the detector and the image signal processing circuits, readout circuit is a critical component in the infrared imaging system. With the development of the CMOS technology, the readout circuit is now moving into the CMOS technology. With the feature size scaling down, the readout cell size is reduced, which enable us to integrate more complex circuits into the readout cell. From the system point of view, different requirements and specifications for the CMOS readout circuit are analyzed and discussed. Different readout circuit parameters such as injection efficiency, dynamic range, noise, detector biasing control, power consumption, unit cell area, etc are discussed in details. Performance specifications of different readout cell structures are summarized and compared. Based on the current mirroring integration readout cell, a fully differential readout cell is proposed. The injection efficiency of this proposed readout cell is very close to unity and the detector biasing voltage is close to zero. Moreover, the dynamic range of the proposed readout cell is increased and the rejection on interference is improved because of the fully differential structure. All these are achieved without much power consumption increasing. Finally, a full digital readout circuit concept is introduced. By employing a current controlled oscillator, the photocurrent is converted to frequency and integrated in digital domain and the final output is digital signal.

Keywords: CMOS, readout circuit, infrared detector, focal plane array

1. INTRODUCTION

The infrared imaging system is widely used in many applications in military, medical and industry. The infrared focal plane array (IRFPA) is the key of the entire infrared imaging system and it determines the overall system performance. The main structure of the hybrid focal plane array is the photon detector array and the readout integrated circuit (ROIC). Each unit cell of the detector array is coupled to the readout cell by means of flip-chip bonding. The main purpose of the readout cell is to extract the photocurrent from the detector cell and process the signal. Low-power, small silicon area and high-performance readout cell is the main concern of the entire FPA design.

The rapid development of the CMOS technologies has made it a very attractive candidate for the IRFPA readout circuit. On IRFPA, the pixel size limits the circuit complexity and hence the performance of the ROIC. With the CMOS feature size scaling-down, it is possible to integrate more signal processing functions into the ROIC within the pixel size and at the same time reduce the power consumption. An even better solution is to integrate the readout circuit as well as the pixel level analog to digital converter (ADC) into the ROIC, making a smart pixel IRFPA. The performance, reliability and flexibility of such smart pixel IRFPA will be greatly increased.

In this paper, different traditional CMOS ROIC structures is reviewed. Performance specifications such as injection efficiency, detector bias control, cell area and power consumption of several readout cell structures are summarized. Based on the specification requirement, a fully differential readout cell structure is then proposed. The fully differential readout structure can increase the dynamic range and improve the rejection on interferences hence increases the signal to

noise ratio of the system. Lastly, a novel full digital readout cell structure with on-chip ADC is presented. This ROIC implements the signal integration in the digital domain and the output is digital signal.

2. READOUT CELL CIRCUIT

2.1 Readout cell circuit and the macro model of the photovoltaic detector

The photovoltaic detector cell used in IRFPAs is the photodiode whose output signal is in current form. As shown in Fig. 1, the photon excited current generated by the photodiode is coupled to the readout cell where the photocurrent charges the integration capacitor and transfer to voltage signal, which is then multiplexed to the final output. The function of the readout cell circuit is to extract the photocurrent from the photodiode and integrate into voltage output. At the same time, the readout cell circuit also provides biasing voltage to the photodiode. For the detector itself, it sees the input resistance of the readout cell R_{in} and certain bias voltage V_{DET} , as shown in Fig.1.

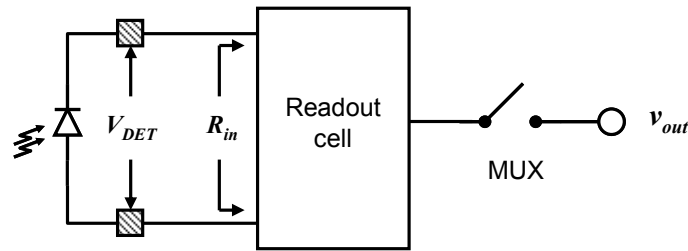


Fig. 1. Infrared photovoltaic detector and readout cell circuit.

Fig. 2 shows the equivalent circuit of the photodiode. The photodiode is modeled as a current source together with a parasitic capacitance C_D and parasitic resistance $R_{\theta}A$. The photon excited current is modeled as the photocurrent i_p and the parasitic capacitance C_D includes all the p-n junction capacitance as well as the bonding capacitance. The shunt resistance is expressed as $R_{\theta}A$, where R_{θ} is the shunt resistance normalized to the detector area under zero biasing voltage and A is the detector area.

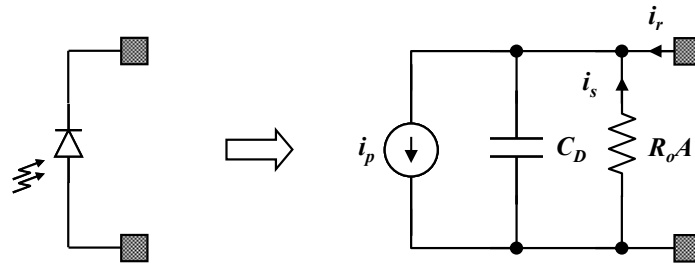


Fig. 2. Infrared photovoltaic detector and its equivalent circuit.

2.2 Performance specifications of the readout cell circuit

As an interface between the photodiode and the multiplexer circuit, the readout cell plays an important role in the entire system. From Fig. 2 it is clear that the photocurrent generated by the photodiode will be shunt partly by the shunt resistance $R_{\theta}A$ and the existence of $R_{\theta}A$ will decrease the photocurrent transferred to the readout circuit. Assume the current i_s is shunt by the resistance $R_{\theta}A$, then the injection efficiency of the readout circuit can be defined as the ratio of the current flowing into the readout circuit i_r to the total photocurrent i_p , as shown bellow:

$$\eta_i = \frac{i_r}{i_p} = \frac{R_0 A}{R_0 A + R_{in}} \quad (1)$$

where η_i is the injection efficiency and the R_{in} is the input resistance of the readout circuit. The injection efficiency is the measure of how much photocurrent is transferred to the readout circuit. For readout circuit, high injection efficiency is preferred so the input resistance of the readout circuit should be significantly lower than the photodiode shunt resistance $R_0 A$.

On the other hand, the parasitic capacitance C_D limits the bandwidth of the readout circuit and it will affect the high-frequency response of the system. As the low input resistance is required for the readout circuit, the bandwidth is normally better than the requirement for normal frame rate systems. For better performance of the imaging system, high $R_0 A$ and low C_D are preferred for the photodiode. However, due to the detector material and process imperfection, the $R_0 A$ is limited. As a result, the input resistance of the readout circuit is required even lower to compensate the limited shunt resistance $R_0 A$.

The dark current, noise and responsivity of the photodiode are affected by the biasing voltage V_{DET} . Ideally the photodiode is biased under zero voltage to maximize the signal to noise ratio of the photocurrent and minimize the biasing power consumption. The readout cell circuit should be able provide a constant zero biasing voltage to the photodiode to obtain optimized performance.

Another performance specification is the dynamic range of the readout circuit. From the system point of view the dynamic range should be high to accommodate drastically varying image contrast. In the traditional readout circuits, the photon excited carriers are accumulated in the integration capacitors and transferred to voltage output signal. Hence the dark current and the integration capacitor determine the dynamic range of the readout circuit. High dynamic range calls for bigger integration capacitor which leads to larger readout cell size.

Readout cell size is another important specification to consider. The readout cell size must be smaller than the detector pixel size to make the array feasible. As the IRFPA image resolution increases, the pixel size is reduced. On the other hand the feature size scaling down of the CMOS technologies makes it possible to integrated more devices into the readout cell.

Power consumption is very important for readout cell circuits. The pixel number of the IRFPA is increased continuously hence the total power consumption of the entire readout circuit is increased as well, which imposes a heavy load for the cooling system.

2.3 Review on the current readout cell circuits

The readout cell is the circuit directly connected to the detector and provides interface to the detector. Some commonly used readout cell circuits are summarized in this section.

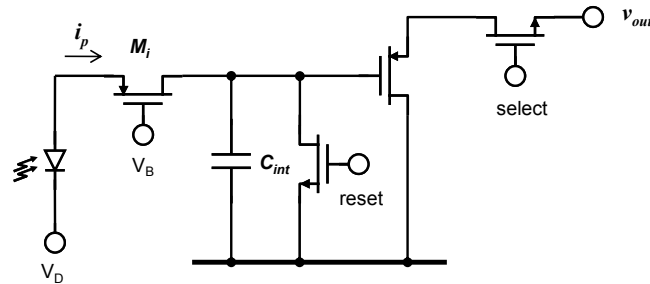


Fig. 3. Schematic of the direct injection (DI) readout cell circuit.

The first simple readout cell is the **direct injection (DI) readout cell** [2] as shown in Fig.3. This is the most basic readout cell as it does not involve any signal processing inside the cell. **The photocurrent is directly injected into the integration**

capacitor C_{int} through a current buffer transistor M_i . Although the photodiode is connected to the source of the current buffering transistor, however the input resistance the photodiode sees is still very high since the drain of the M_i is a high resistance node. Consequently the injection efficiency of the DI cell is very low. Moreover, as the biasing of the photodiode is obtained by tracking the V_B with the V_{GS} of transistor M_i , it is difficult to provide stable biasing as global matching of the transistor threshold voltage is poor. As a result, the biasing of the photodiode is not precisely close to zero. Furthermore the DI cell provides no current the photocurrent.

The gate modulation input (GMI) readout cell circuit [3] is shown in Fig. 4. It solves the high input resistance problem of the DI cell and provides current gain. The input resistance is now reduced to $1/g_{mi}$. However, since the photocurrent flowing through M_i is very weak, not much transconductance is generated. The value of the input resistance $1/g_{mi}$ is not low as a consequence. The injection efficiency is improved compared to DI cell, but is not very high. The biasing problem remains the same as the DI cell since the tracking of V_D to V_{GS} globally is not precise. One of the advantages of the GMI cell is that it provides current gain without active power consumption.

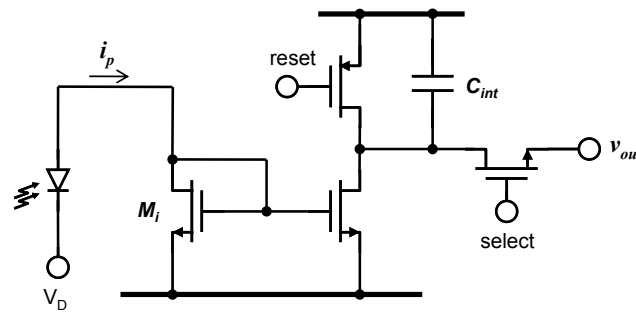


Fig. 4. Schematic of the gate modulation input (GMI) readout cell circuit.

The buffered direct injection (BDI) readout cell [4], as shown in Fig. 5, has lower input resistance and can provide precise biasing voltage to the photodiode compared to the DI readout cell. An amplifier with gain of A_v is adopted to provide negative feedback to the gate of the transistor M_i . As a result, the input resistance of the BDI cell is reduced by a factor of A_v compared to DI cell and the injection efficiency is increased to close unity. The virtual short property of the amplifier makes the biasing of the photodiode stable. By simply provide the same voltage on V_B and V_D , the biasing voltage across the photodiode is zero with an error of the amplifier input offset voltage, which can be designed very low. Apparently the drawback of the BDI cell is that an additional amplifier is required, which causes the active power consumption and increases the cell area and additional noise. To save power and area, half of the amplifier circuit can be shared by all the readout cells [1].

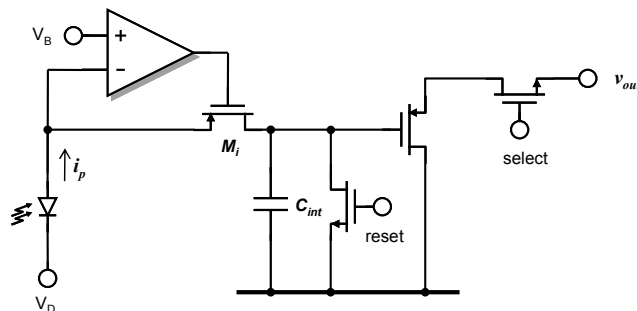


Fig. 5. Schematic of the buffered direct injection (BDI) readout cell circuit.

Another readout cell with active amplifier involved is the capacitive transimpedance amplifier (CTI) readout cell [5], as shown in Fig. 6. It can also provide lower input resistance and precise biasing voltage to the photodiode similarly to the BDI cell. The drawbacks of the CTI cell are the active power consumption, additional noise and no current gain.

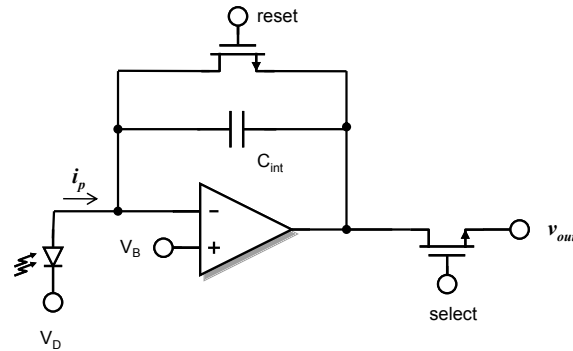


Fig. 6. Schematic of the capacitive transimpedance amplifier (CTI) readout cell circuit.

The buffered gate modulation input (BGMI) readout cell [6] is shown in Fig. 7. It is similar to the buffered direct injection cell by adopting an amplifier to buffer the input photocurrent and achieve very low input resistance, hence has close to unity injection efficiency. Similar to the GMI cell, the current mirror can provide current gain and make the background suppression easier.

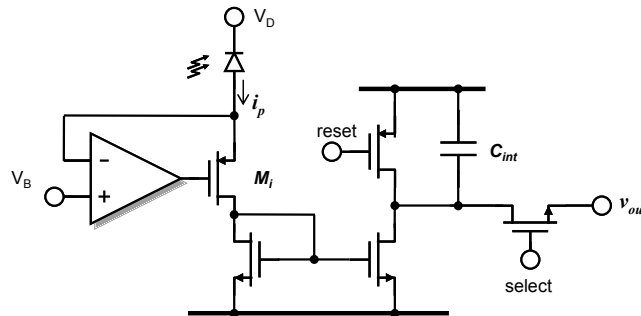


Fig. 7. Schematic of the buffered gate modulation input readout cell.

For all the readout cells mentioned above, we can conclude that the main approach to achieve low input resistance and stable zero biasing is to adopt an active amplifier. However, the amplifier involves active power consumption and larger cell size. Often the amplifier contributes more noise as well. For high resolution IRFPAs, a small increase on the cell power consumption will result in high power consumption as there are many cells in the IRFPA. This is the main burden of using the above mentioned high performance readout cells in high resolution IRFPA systems.

The current mirroring integration (CMI) readout cell [7] is shown in Fig. 8. It solves the input resistance and detector biasing problem without using any active amplifier. It uses a self-biased current source consists of a pair of current mirrors. Transistor M_{n1} and M_{n2} are equally sized to form a unity current mirror, while transistor M_{p1} and M_{p2} are equally sized to form another unity current mirror. These two current mirrors are coupled to form a self-biased current source. In terms of the input resistance, detect biasing control, noise performance, power consumption and cell size, the CMI cell offers very good performance. Firstly, the input resistance the detector seen is very low. By doing smaller signal analysis on the cell circuit, the input resistance the detector seen is expressed as ratios of different transistor transconductance values, as shown in (2).

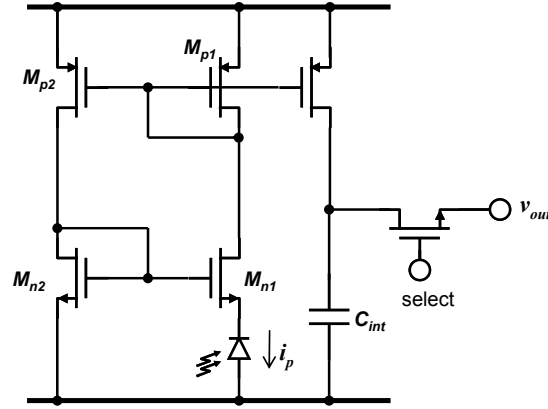


Fig. 8. Schematic of the current mirroring integration (CMI) readout cell.

$$R_{in} = \frac{1}{g_{mn2}} \left(1 - \frac{g_{mp1} g_{mn2}}{g_{mn1} g_{mp2}} \right) \quad (2)$$

It is clear that for perfect matched transistors, the input resistance is ideally zero. Even when mismatch is considered, the input resistance is still very low compared to other types of readout cells. This low input resistance improves the injection efficiency to close unity without using any active amplifier. The detector biasing voltage V_{DET} is given by

$$V_{DET} = \frac{K_p}{K_n} \Delta V_{THp} + \Delta V_{THn} \quad (3)$$

where K_n and K_p are transistor current factor of NMOS and PMOS transistors, respectively, ΔV_{THn} and ΔV_{THp} are threshold voltage mismatch of NMOS and PMOS transistors, respectively. It is seen that both the input resistance and the detector biasing voltage are determined by the matching of the transistors. By properly design the transistor size, the mismatch of the transconductance and threshold voltage can be reduced to less than 1% as these transistors are closed each other. The resulting injection efficiency is very close to unity and the detector biasing voltage is close to zero. The extracted photocurrent can be easily mirrored out for integration. The background suppressing circuit can be easily applied to the CMI cell as well. The CMI cell offers very attractive performance compared to other readout cells. The close to unity injection efficiency and close to zero biasing voltage greatly improve the uniformity of the entire IRFPA. To further improve the performance, cascoded current mirrors can be used in the CMIS cell.

In this section, different readout cells are introduced and their performances are compared. A more detailed comparison among all the readout cells can be found in [8].

2.4 Proposed fully differential current mirroring integration readout cell circuits

Due to the symmetric property of the CMI cell, it is easy to implement the fully differential readout cell immune to cross talk and interference. The proposed fully differential current mirroring integration (DCMI) readout cell is shown in Fig. 9. The extracted photocurrent is mirrored out by two current mirrors, generating two signal currents i_{pp} and i_{pn} . These two signal currents i_{pp} and i_{pn} are integrated into two integration capacitors and produce the differential output $v_{oup} - v_{outn}$. Since the differential nature of the output signal, the cross-talk, charge injection due to the multiplexer switches and interference are treated as common mode signals and are rejected. Besides the rejection to the cross-talk and interference, the dynamic range of the readout cell is also doubled, which is very attractive in advanced CMOS technologies. All these are gained by little cost. The properties of the DCMI cell greatly improve the uniformity and robustness of the practical IRFPAs. Due to the high performance and low power consumption properties of the proposed DCMI readout cell, it can be widely used in high performance high resolution IRFPA systems.

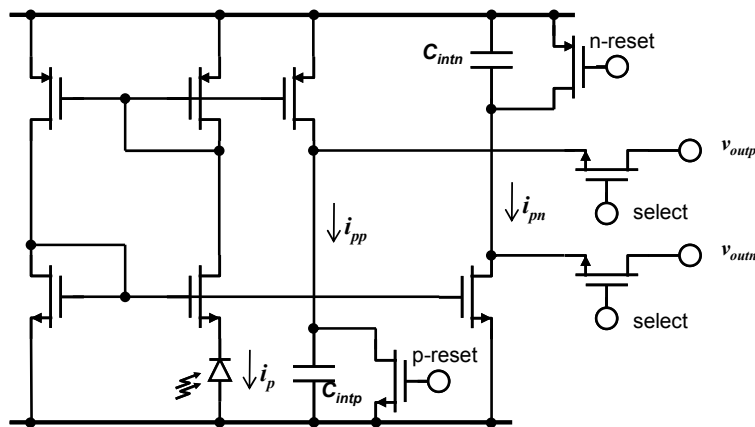


Fig. 9. Schematic of the fully differential current mirroring integration (DCMI) readout cell.

3. FULL DIGITAL READOUT CIRCUIT

3.1 Proposed ROIC with off-cell digital integration circuit and column ADC

The output of the traditional ROIC is pure analog signal. For infrared imaging systems the output of the ROIC will be converted into digital signal for digital signal processing. It is desired to convert the analog signal into digital signal inside the ROIC for better signal transfer and signal to noise ratio of the entire system. The key feature of the on-chip analog to digital converter (ADC) is the low-power consumption high dynamic range. In this section, a ROIC with on-chip digital integration circuit and column ADCs is introduced, as shown in Fig. 10.

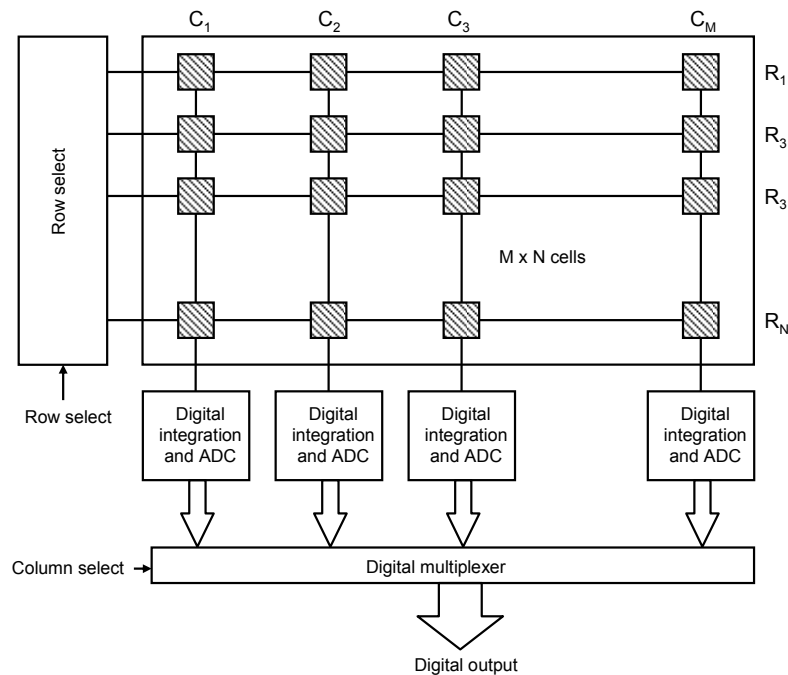


Fig. 10. Block diagram of the proposed ROIC with on-chip column ADCs.

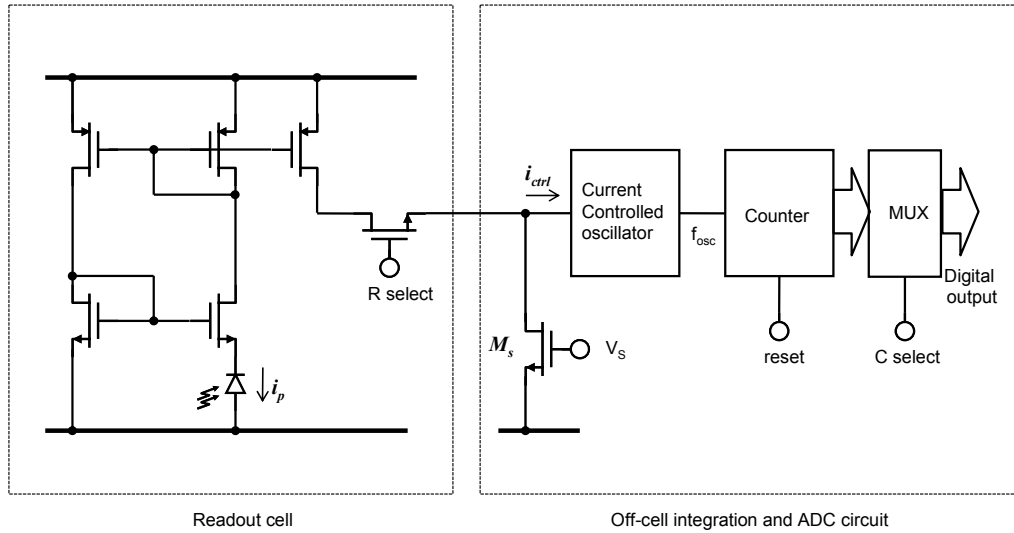


Fig. 11. Schematic of the readout cell, the digital integration circuit and ADC.

As shown in Fig. 10 the ROIC consists of the $M \times N$ readout cells and M digital integration circuit and ADC blocks. Each block includes a current controlled oscillator (ICO) and a counter. The basic conceptual schematic is shown in Fig. 11. The photocurrent is extracted by the CMI cell and mirrored out as i_{ctrl} . After the row select multiplexer, the photocurrent controls an ICO and the output pulse number is accumulated in a counter. In certain frame period, the counted value is the pulse number, which is proportional to the photocurrent. Finally the counter value is the digital output signal. Transistor M_s is used to implement the background suppression. By using this digital integration circuit, the integration capacitor is removed, and hence the readout cell size is reduced. The drawback of this scheme is the integration time is not a whole frame time, but a row scanning time, which will degrade the sensitivity of the system. One possible solution is the hybrid integration scheme that keeps a small integration capacitor inside the read cell. By combine the integration in both analog domain and digital domain, full time integration is achieved.

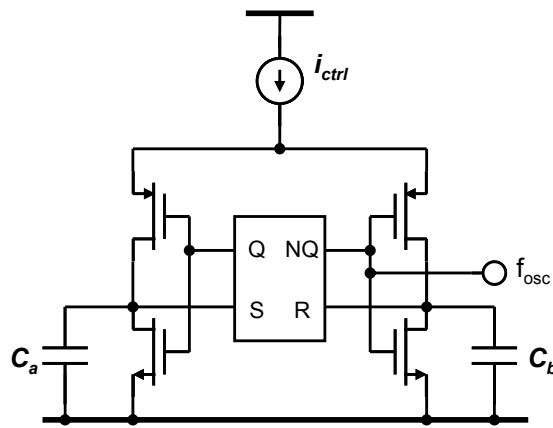


Fig. 12. Schematic of the current controlled relaxing oscillator.

The key component of this scheme is the ICO whose linearity and dynamic range will determine the system performance. The proposed ICO schematic is shown in Fig. 12. The period of the ICO output signal is composed of capacitor charging times t_c and RS latch delay t_d . The control current i_{ctrl} is switched to charge capacitors C_a and C_b alternatively, hence the period of output signal can be defined as $T=2(t_c + t_d)$. The output frequency of the ICO is:

$$f_{osc} = \frac{1}{T} = \frac{1}{2(t_c + t_d)} = \frac{1}{2(\frac{C \times V_{th}}{i_{ctrl}} + t_d)} = \frac{i_{ctrl}}{2(C \times V_{th} + t_d \times i_{ctrl})} \quad (4)$$

where $C=C_a=C_b$ and V_{th} is threshold voltage of the RS latch. It can be seen that the ICO output frequency is not a pure linear function of the control current. One can minimize the t_d to increase the linearity of the ICO. On the other hand, as the frequency and current relationship is determined in (4), it is possible to compensate the non-linearity in the digital domain. If non-linearity is not a concern, this ICO can achieve a very large dynamic range yet consumes dynamic power only, making it a good candidate for the ROIC integration circuit and ADC.

3.2 Simulation result on the ICO

The main properties of the ADC in the readout circuit are the dynamic range and the linearity. For the proposed ADC the linearity and dynamic range is mainly determined by the ICO. The relaxing oscillator exhibits very good performance. Simulated using a 0.35- μm CMOS technology file, the frequency vs. the control current curve and the linearity are illustrated in Fig. 13. It is seen the dynamic range of the proposed ICO is more than 80 dB and the linearity error is less than 0.2%. The dynamic range can be extended further to more than 120 dB at a cost of worse linearity. However this is not a big problem as anyway the Gamma correction will be performed later and compensation scheme can be implemented in the digital domain.

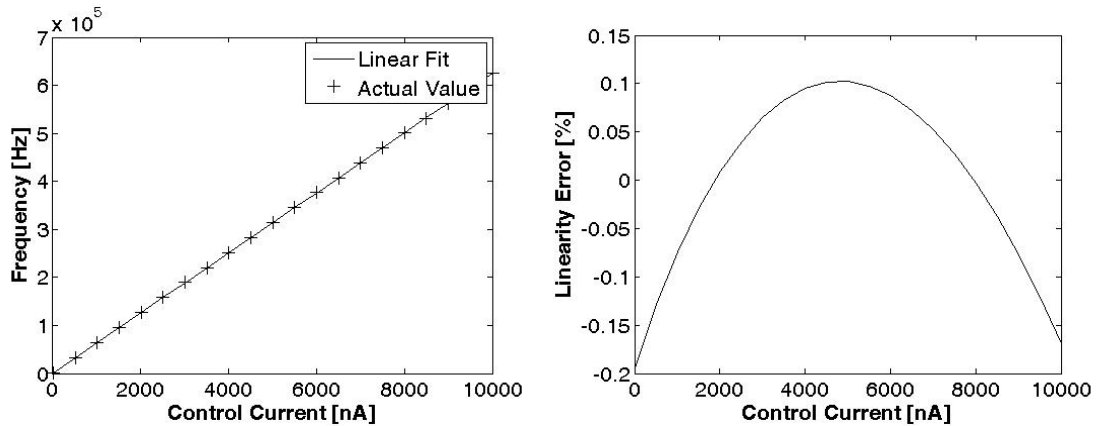


Fig. 13. Simulated frequency vs. control current curve (left) and linearity error (right) of the current controlled oscillator.

4. CONCLUSIONS

In this paper, the basic requirements on ROIC for infrared detector are discussed. Different readout cells are reviewed and their strengths and weaknesses are discussed. Based on the CMI readout cell, a fully differential readout cell is proposed. Besides the close to unity injection efficiency and the close to zero detector biasing voltage, the dynamic range of the proposed readout cell is increased, which is important for imaging systems. On the other hand, the rejection to cross talk, charger injection from the multiplexer switches and other interference is greatly improved because of the fully differential structure. By using the CMI readout cell, a full digital readout circuit concept is introduced. The proposed ROIC includes a current controlled oscillator to convert the photocurrent into frequency. Counted by a digital counter, the photocurrent signal is integrated in digital domain and output the digital signal. All these functions are done in digital domain and consume dynamic power only. The dynamic range and linearity of the proposed ICO is suitable for this application. With the development of the CMOS technology, the power consumption and area of the proposed digital readout structure will be decreased, making this structure a more attractive solution for CMOS imager sensor ROICs.

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